Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	Summer	**Academic Year**	2024/2025

Course Details

Course Title	System Analysis and Design			**Course No.**	0412401		
Type of Learning	☐ Blende						
Credit Hours	3	**Theoretical**	3	**Practical**	0		
Course Level (according to JNQF)			7	**JNQF Hours***	12		
Pre-requisite	Object-Oriented Programming C++ - 0411203			**Co-requisite**	N/A		

Course Instructor Information

Instructor Name	Dr. Shadi Ettantawi		
Instructor E-mail	s.ettantawi@meu.edu.jo		
Course Time	S M T W 12:30 – 2:00		
Office Hours	S M: 11:00 - 12:00		
Office Number	N142	**Office Phone**	N/A
Lab Instructor Name –If assigned-	N/A		

Sources and References

1. Textbook

System Analysis & Design in a Changing World by John W. Satzinger, Robert B. Jackson and Stephen D. Burd, 6 Edition, 2012. ISBN 1-111-53415-2.

2. References

- Modern Systems Analysis and Design, J. S. Valacich, Joey F. George and Jeffrey A. Hoffer, Published by Pearson, 9th Edition, September 18th 2020
- Systems Analysis and Design, 11th Edition, Scott Tilley and Harry J. Rosenblatt, 2017, Print ISBN: 9781305494602

Teaching Methods

1. Lectures/ Discussion	**2.** Case Studies	**3.** Active Learning
4. Guest Speakers	**5.** Projects/ Presentations	

Course Description

This course provides students with an in-depth understanding of the different phases application/system development, covering topics such as specifying what application/system should do, requirements gathering techniques, designi application/system architecture, implementing the components of the application/syste and using structured and object-oriented models to represent system functions and d structures. Additionally, the course covers topics related to human-computer interfa database design, and project management.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLO and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Compete
1	Describe the information systems development life cycle (SDLC).	PLO 1: Analyze a complex computing problem and apply principles of computing to identify solutions.	K1-I		
2	Apply project management on system design and evaluate system development methodologies.	PLO 5: Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		S2-D	
3	Discuss the modern approach to system requirements, analysis, and design that combines both process and data views of systems.	PLO 3: Communicate effectively in a variety of professional contexts.	K1-P		
4	Illustrate the use of traditional (structured) and object-oriented (OO) approaches to systems analysis and design	PLO 1: Analyze a complex computing problem and apply principles of computing to identify solutions		S1-D	
5	Discuss methodologies of designing systems database, forms, reports, and interfaces.	PLO 2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	K1-P		

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summat Format
1	- Syllabus discussion - Ch1: Introduction - SDLC - Requirements	Lecture (1, 2, 3,4)	Lecture	1		
2	- Stakeholders - Information Gathering		Lecture	1		
	- Use-case Modeling		Lecture	3		
3	- **Quiz #1.** - Use-case Modeling - Domain modelling		Lecture	3, 4	Quiz 1	Summa
4	**Midterm Exam**				Midterm Exam	Summat
5	- Designing Database		Lecture	3, 4		
6	- **Quiz #2.** - **Project Submission.** - System Sequence Diagram - Activity Diagram		Lecture	4, 5		
7	- Project Management - Designing user interface		Lecture	4, 5	Quiz 2	Summat
8	**Final Exam**				Final Exam	Summat

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 4
2 Quizzes	20%	Week 3,6
Project	15%	Week 6
Final Exam	40%	Week 8

Course Policies

Item	Policy
Quizzes	▪ No makeup quizzes. ▪ A minimum of 2 quizzes are given. ▪ If more quizzes are given, then the lowest quiz's grade is dropped.
Exams	▪ The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.

Makeup Exams	▪ Makeup exam should not be given unless there is a valid excuse.
Drop Date	▪ The last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exams or quizzes is an illegal and unethical activity ▪ Standard MEU policy will be applied. ▪ All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. ▪ Sign-in sheets will be circulated. ▪ If you miss class, it is your responsibility to find out about any announceme or assignments you may have missed.
E-Learning	▪ It is your responsibility to check the course's eLearning website regularly (a least once every 2-3 days) for announcements and material.
Workload	▪ The student should expect to spend 6 hours per week as an average workloa
Graded Exams	▪ Instructor should return exam papers graded to students within one week aft the exam date.
Participation	▪ Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. ▪ Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Assignments are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team wi give at least one presentation regarding a selected case, delivering a fully working application. ▪ More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedback (Project)

Criteria	Excellent	Very Good	Adequate	Poor
Initiate the project and create a project plan	Excellent understanding of the project plan, analysis of the problem, clear definition of actors, and clearly documented functional and non-functional requirements. (5 Marks)	Good understanding of the project plan, analysis of the problem, clear definition of actors, and clearly documented functional and non-functional requirements. (4 Marks)	Adequate understanding of the project plan, analysis of the problem, clear definition of actors, and clearly documented functional and non-functional requirements. (3 Marks)	Poor understanding of the project plan, analysis of the problem, clear definition of actors and clearly documented functional and non functional requirements. (2 Marks)
Analyze project requirements and develop a project specification document (SRS)	Clearly defining main activities, effective application of modeling tools to perform data modeling such as	Good defining main activities, effective application of modeling tools to perform data modeling such as	Adequate defining of main activities, effective application of modeling tools to perform data modeling such as	Poor defining mair activities, effective application of modeling tools to perform data modeling such as

	Use-Case and ERD, Effective use of relations to capture interactions. (5 Marks)	Use-Case and ERD, Effective use of relations to capture interactions. (4 Marks)	Use-Case and ERD, Effective use of relations to capture interactions. (3 Marks)	Use-Case and ERD Effective use of relations to capture interactions. (2 Marks)
Design the project in alignment with project requirements and develop a project design document (SDS)	Clearly designing the software architecture such as mobile, or web architecture, Well-designed front-end (interface), and back-end (reporting details). (5 Marks)	Good designing software architecture such as mobile, or web architecture, Well-designed front-end (interface), and back-end (reporting details). (4 Marks)	Adequate design the software architecture such as mobile or web architecture, Well-designed front-end (interface), and back-end (reporting details). (3 Marks)	Poor designing the software architecture such a mobile or web architecture, Well-designed front-end (interface), and back-end (reportin details). (2 Marks)

* **JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3 hours	14	42
Tutorials	0	0	0
Laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1 hours	1	1
Second Exam (If applicable)	0	0	0
Final Exam	2 hours	1	2
Quizzes	0.5 hour	2	1
Self-Learning Activities			
Assignments/ Homework	3 hours	6	18
Case Studies	0	0	0
Presentations	1 hour	1	1
E-learning	0	0	0
Extra Readings	4	14	56
Total Hours	**121**		
JNQF Hours	**121/10 = 12.1**		